

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: PRINCIPLES OF OPERATING SYSTEM

Practical No.7

Aim:

Process Synchronization and Bounded Buffer Problem

- Simulate producer-consumer bounded buffer using mutex and semaphores.
- Implement buffer control with synchronized access.
- Introduce circular queue techniques for managing shared buffers.

Code:

```
import threading
```

```
import time
```

```
BUFFER_SIZE = 5
```

```
buffer = [None] * BUFFER_SIZE
```

```
in_index = 0
```

```
out_index = 0
```

```
mutex = threading.Semaphore(1)
```

```
empty = threading.Semaphore(BUFFER_SIZE)
```

```
full = threading.Semaphore(0)
```

```
def producer():
```

```
    global in_index
```

```
    for item in range(1, 11):
```

```
        empty.acquire()
```

```
        mutex.acquire()
```

```
        buffer[in_index] = item
```

```
        print(f"Produced: {item}")
```

```
        in_index = (in_index + 1) % BUFFER_SIZE
```

```
        mutex.release()
```

```
        full.release()
```

```
        time.sleep(0.5)
```

```
def consumer():
```

```
    global out_index
```

```
    for _ in range(10):
```

```
        full.acquire()
```

```
        mutex.acquire()
```

```
        item = buffer[out_index]
```

```
        buffer[out_index] = None
```

```
        print(f"Consumed: {item}")
```

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```
out_index = (out_index + 1) % BUFFER_SIZE

mutex.release()
empty.release()
time.sleep(1)

producer_thread = threading.Thread(target=producer)
consumer_thread = threading.Thread(target=consumer)

producer_thread.start()
consumer_thread.start()

producer_thread.join()
consumer_thread.join()

print("Process completed successfully.")
```

Output:

```
C:\Deepak\OS>python Practical_7.py
Produced: 1
Consumed: 1
Produced: 2
Consumed: 2
Produced: 3
Produced: 4
Consumed: 3
Produced: 5
Produced: 6
Consumed: 4
Produced: 7
Produced: 8
Consumed: 5
Produced: 9
Produced: 10
Consumed: 6
Consumed: 7
Consumed: 8
Consumed: 9
Consumed: 10
Process completed successfully.
```